


# Sunny Yadav

UG (III Year I Semester)
B.Tech. (Electronics & Communication Engineering)
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Indian Institute of Technology Roorkee

**VERIFIED**


## Area of Interest

Software Engineering, Machine Learning, Generative AI

## Education

| Year | Degree/Examination | Institution/Board | CGPA/ Percentage |
|---|---|---|---|
| 2026 | B.Tech. 2nd Year | Indian Institute of Technology, Roorkee | 8.456 |
| 2024 | Intermediate (Class XII) | SUDITI GLOBAL ACAD VILL NAGLA PO DHANUAN ETAWAH UP (CBSE) | 95.20 % |
| 2022 | Matriculate (Class X) | ST PETER`S SCH KESTH PO JASWANTHNGR ETAWAH UP (CBSE) | 97.20 % |

## Projects

**PCB-based high frequency signal receiver** | IIT ROORKEE — September 2025 - November 2025

- Fabricated a 2-layer RF receiver PCB integrating amplification and filtering stages for high-frequency signal reception.
- Performed simulation, PCB layout, and hardware validation using oscilloscope and spectrum analyzer measurements.
- Tech Stack: PCB Design, RF Electronics, Oscilloscope, Spectrum Analyzer

**PLACEDESK:Automated Placement Management Platform** | ArIES, IIT ROORKEE — October 2025 - November 2025

- Developed a placement management platform with role-based auth (student/admin) and secure notifications.
- Engineered a Llama-3.3-70B pipeline for automated email generation, classification, and summarization.
- Deployed a RAG chatbot and noticeboard surfacing verified eligibility and company schedules.

**Traffic light using Counter** | IIT ROORKEE — March 2025 - April 2025

- Designed a 4-way traffic light controller circuit using a 555 timer IC and CD4017 decade counter.
- Configured the 555 timer in astable mode to generate clock pulses driving sequential counter states.
- Tuned RC timing networks and diode-based output logic to sequence Red-Yellow-Green phases accurately.

**Memory Management Simulator** | ArIES, IIT ROORKEE — May 2026 - June 2026

- Built a C++ OS memory simulator modeling an allocator, Buddy System, MMU, and cache hierarchy.
- Implemented First/Best/Worst Fit allocation and an O(1) XOR-based Buddy allocator.
- Modeled L1 FIFO/L2 LRU caches with MMU-driven page-faults and virtual address translation.

**Single-Cycle RISC-V (RV32I Base) Microprocessor in Logisim** | IIT Roorkee — April 2026

- Developed a 32-bit single-cycle RISC-V microprocessor executing the RV32I Base Integer ISA.
- Architected the datapath and Control Unit to execute R, I, S, B, U, and J-type instructions.
- Designed modular hardware components including a Register File, ALU, and isolated memory units.

**Schizophrenia Detection via EEG** | ArIES, IIT ROORKEE — February 2026 - March 2026

- Converted raw EEG time-series into 2D spectrograms to classify subjects as Schizophrenia patients or healthy controls.
- Trained VGG16, ResNet50, and ViT-B/16 classifiers, then combined them via a soft-voting ensemble.
- Achieved 78% accuracy (ResNet50) and 0.90 patient recall with the ensemble, minimizing missed diagnoses.

**Single-Cycle RISC-V (RV32I Base) Microprocessor in Verilog** | IIT Roorkee — March 2026

- Developed a 32-bit RISC-V microprocessor in Verilog simulating the RV32I Base Integer ISA to understand low-level code execution.
- Architected the core datapath and Control Unit at the Register-Transfer Level (RTL) to decode and execute R, I, S, B, U, and J-type instructions.
- Designed modular Verilog components (ALU, Register File, and memory units) to manage data retrieval and program flow.

**HackHunt: RAG-Based Hackathon Aggregator** | ArIES, IIT ROORKEE — May 2025 - June 2025

- Built a platform scraping Devfolio, Unstop, and HackerEarth, aggregating 60-80 hackathons, deduplicated via MongoDB.
- Built a RAG chatbot using FAISS vector search (Sentence Transformers) and local LLaMA 3.2 inference via Ollama.
- Deployed Chainlit interface, letting users query hackathons in natural language instead of manual browsing.

**Yolo to DETR computer vision** | ArIES, IIT ROORKEE — June 2025 - July 2025

- Built a custom PyTorch pipeline converting YOLO-format annotations to COCO-style boxes for DETR.
- Fine-tuned a pretrained DETR (ResNet-50 backbone) via HuggingFace across 13 object classes.
- Trained with mixed precision and differential learning rates, reaching a 1.22 validation loss.

**Sahayak: Personalized Learning & Revision Assistant** | ArIES, IIT ROORKEE — September 2025 - October 2025

- Built a Streamlit platform generating personalized quizzes and Q&A for Class 6-10 Science from NCERT content.
- Built a LangChain RAG pipeline with vector embeddings for curriculum retrieval, using Ollama for question generation.
- Designed an adaptive evaluation engine that auto-grades responses and adjusts difficulty based on performance.

**Acoustic Waveform Signal Recovery** | ArIES, IIT ROORKEE February 2026 - March 2026

- Built a Bi-LSTM model in PyTorch to reconstruct missing points in 80,000 sparse audio sequences.
- Designed a masked MSE loss and single-pass amortized inference matching the competition's constraints.
- Trained for 20 epochs, achieving 0.0020 validation MSE, then deployed a batched GPU inference pipeline.

**Multi-Modal House Price Prediction** | ArIES, IIT ROORKEE December 2025 - January 2026

- Built a multimodal pipeline fusing tabular housing data with satellite imagery for neighborhood context.
- Extracted 512-dim ResNet-18 embeddings, reduced via PCA, fused with tabular features in XGBoost.
- Achieved an R^2 of 0.900, outperforming baselines; validated spatial features via Grad-CAM.

## Awards / Scholarships / Academic Achievements

- Global Rank 1740 | Codeforces Round 1071
- Global Rank 496 | Codeforces Round 1090
- Global Rank 980 | Codeforces Round 1072
- Global Rank 1552 | Codeforces Round 1089
- Global Rank 238 | Codeforces Round 1080
- Global Rank 1082 | Codeforces Round 1102
- Global Rank 1331 | Codeforces Round 1095
- Global Rank 506 |Educational Codeforces Round 191
- Rank 3 in "Overall Excellence Award" at Srishti 2025 as part of Team ArIES.
- Global Rank 2870 | Codeforces Round 1091

## Skills

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|---|---|
| Computer languages | C++,Python |
| Software Packages | Git, Jupyter Notebook, Visual Studio Code, MATLAB, Pandas, Pytorch, Numpy |
| Languages Known | English, Hindi |

## Positions of Responsibility & Extra Curriculars

**AI/ML Sub-Head** | ArIES, IIT Roorkee May 2026 - Present

- Contributed to ML/DL projects and competed in hackathons.
- Organized Neural-Craft 2026, managing execution and registrations.

**Volunteer** | NSS JEE Cell, IIT Roorkee October 2024 - May 2025

- Mentored JEE aspirants by providing targeted strategic guidance and academic support to improve performance.
- Designed comprehensive mock test series for JEE, ensuring strict alignment with current exam patterns.

**Competitive Programming and Organization** | Online Judges October 2024 - Present

- Active member on various online judges including Codeforces,LeetCode.
- Maximum rating on Codeforces: 1668 (Expert) (Handle: sunny_iitr28)

## References

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